Summary of the Project

After the donation of 30+ VR headsets by Honeywell, our group is tasked with the integration of virtual/augmented reality into the FSE100 curriculum at Arizona State University. The FSE100 classes will be targeting SCAI students primarily those who have a Computer Science Major. In stage 1, we finalized that we will be working on Unity and made sure most of the members learned how to use this software.

In stage 2, we will work with FSE and CS faculty to develop a 2-week curriculum that will teach students the basics of VR and how it can be incorporated into engineering. Then, we will design an activity to challenge the students to apply what they learn in a practical setting.
We will work with FSE professors (Steven Osburn) to create several 30-minute lectures and multiple activities surrounding the applications of VR in engineering. The curriculum will eventually teach students how VR technology can be used in Software Engineering, Computer Engineering, Aerospace Engineering, Mechanical Engineering, Civil Engineering, Electrical Engineering, Biomedical Engineering, and more.

Section 1: Project Identification

1.1 Description of the Community Partner



Our community partner is Professor Steven Osburn at The Fulton Schools of Engineering (FSE). The Fulton Schools of Engineering (FSE) is the largest engineering and technology school in the United States, with over 30,000 students enrolled this year. Renowned for its world-class faculty and commitment to student success, FSE has consistently elevated the quality of its programs, fostering innovation and excellence across all levels of education. As the institution continues to grow, it has established a set of core values, cultivating excellence, delivering impactful innovation, encouraging bold thinking, fostering collaboration, and building a foundation for success, to guide its strategies and decision-making. These values not only differentiate FSE within the academic community but also underscore its dedication to creating a culture of learning, collaboration, and meaningful impact.

The goal Professor Steven Osburn has at FSE with this project is to make a four-class session curriculum that uses presentations along with VR experiences to teach students how VR can be used in engineering. The objective of this project is for FSE 100 students to have an opportunity to become familiar with VR technology, learn engineering skills using that technology, and be able to apply those skills in the classroom. The organization's mission is to make concepts taught in FSE 100 easier to learn and understand, and to make activities related to the concepts taught more engaging and interactive.

The key people involved in this organization are:

- Professor Steven Osburn, Community Partner
- FSE 100 Students, Stakeholders

1.2 Needs Assessment

Current State of Our Community Partner

Our community partner is working with us to refine the methods, structure, and implementation of our project. As an FSE100 professor, he provides us with valuable insight into the nature and structure of his classes, and how our curriculum would best fit in. As our plans change, we continue to meet with him to confirm that our project continues to best fit his needs. He has yet to see the first design of our VR project, but as the project develops, plans are made for the community partner to know the current project's state, along with its progress.

Desired Outcomes

Our community partner wants the students to have an opportunity to show the skills they learned through the curriculum through several deliverables. The content of the curriculum was secondary in importance, as the primary concern is for students to have an opportunity to learn a new skill and practice that skill.

Gap Between Desired Outcomes and the Current State

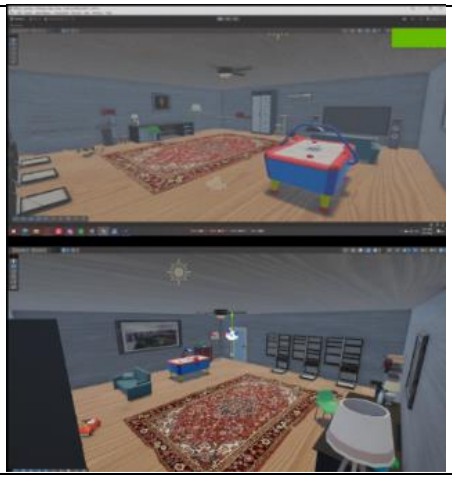
The current state of our project is on track to reach the desired outcome, but there is still a significant amount of work to be done. The curriculum side of our product is proceeding as planned, being developed and iteratively improved to fit the needs of the community partner. However, the technology aspect is still far off, as our team is still learning how to most effectively implement the tools at our disposal.

Forces, Limits, Conditions, Factors

Section 3: Conceptual Design

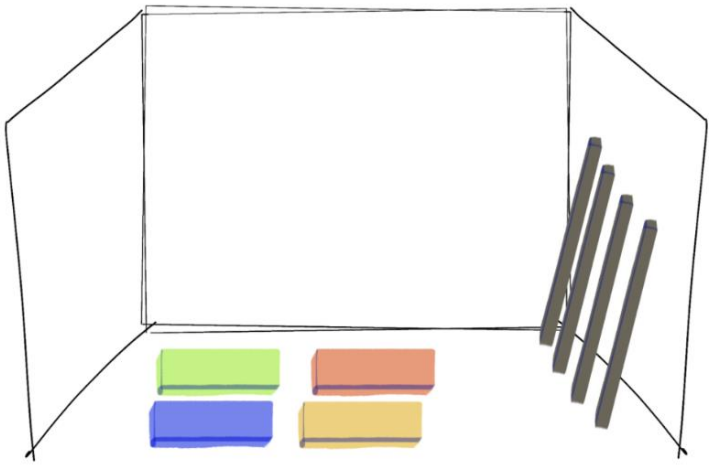
3.1 Proof of Brainstorming

Brainstormed Idea #1 <<SPRING 2025>>

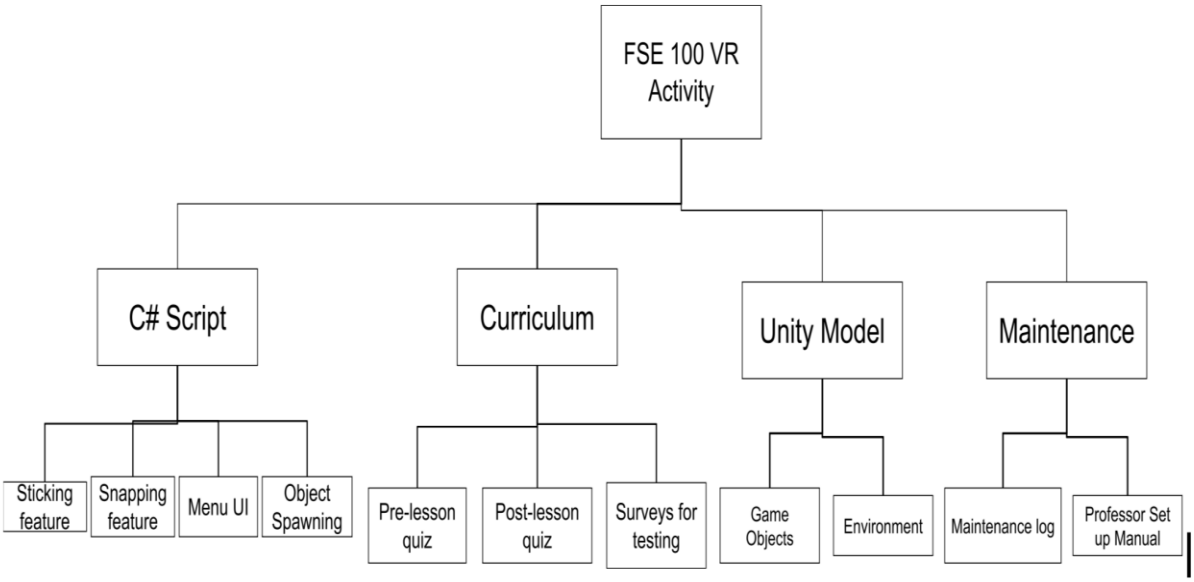
Method Used to Generate Ideas	Research and compilation and design
Number of Concepts Generated	1
Viable Concepts Ideated	1
Documentation of the Ideas Developed (i.e., sketches, bulleted list of solutions, etc.)	
Idea Description	This was our initial brainstormed and modeled out idea for the environment in which the activities would be completed.

Brainstormed Idea #2 <<FALL 2025>>

Method Used to Generate Ideas	Research and compilation and design
Number of Concepts Generated	1
Viable Concepts Ideated	1

Documentation of the Ideas Developed (i.e., sketches, bulleted list of solutions, etc.)	
Idea Description	This was an idea to have a bookshelf put together as an activity. The parts are just around the room, so the students have to pick them up and put the shelf together.

3.2 Functional Decomposition



3.3 Decision Matrix

Decision Matrix – Fall 2025

Final Solution Evaluation		Interactive Block Coding		3D Object Construction		FSE Assembly Line	
Criteria	Weight	Score	Total	Score	Total	Score	Total
Relevance to Curriculum	30%	8	240	7	210	4	120
Ease of Implementation	20%	4	80	8	160	4	80
Student Engagement	30%	6	180	9	270	9	270
Educational Value	20%	8	160	7	140	9	180
	Total		440		**780**		650

Criteria Weight - Fall 2025

Relevance to Curriculum: Fits the standards of the FSE curriculum

Score of 1: Least Relevant to the curriculum

Score of 10: Most Relevant to the Curriculum

Ease of Implementation: Time, training, and support required

Score of 1: extensive prep and a complex setup

Score of 10: ready to start immediately

Student Engagement: Expected student interest and interactivity

Score of 1: passive experience with limited engagement

Score of 10: very immersive, with hands on collaboration

Educational Value

Score of 1: only good to use in one semester and can't expand curriculum

Score of 10: Ability to use over multiple semesters and expand curriculum easily

Product Evaluation Score – Fall 2025

Final Solution Evaluation	Interactive Block Coding	3D Object Construction	FSE Assembly Line
Relevance to Curriculum	It can be very relevant when it comes to computer science principles of the curriculum.	This can be somewhat relevant to the Mechanical Engineering side of the curriculum.	Although taught, it is not something that is relevant and used in current careers. Especially with AI being used.
Ease of Implementation	Requires lab setup and scheduling but can be manageable with TA help	Simple to deliver and easy to give across sections	Complex setup and requires technical work and planning
Student Engagement	Moderate engagement, students have a lot to drag and drop with guidance.	Very interactive and immersive	Very engaging and encourages students to problem solve
Educational Value	Reusable modulus with small updates to the activity	Easily reused over multiple semesters with many objects.	Can definitely be expanded on as it can be used for many types of physical objects.

Conclusion: The 3D Object Construction seems to be the most practical for FSE 100 Classrooms. It sets a foundation that can be scaled to evolve to other areas such as the Interactive Block Coding and introduce more complex training and resources. This will help establish a curriculum that is geared toward student learning and progressing this area of learning

3.4 Final Proposed Solution

This section should include a complete description of the proposed design concept, including sketches, process diagrams, or other artifacts to convey the concept. Consider whether this

solution may be patentable. This solution should be approved by the advisor and partner before proceeding to testing and detailed design.

Complete Description of Proposed Design Concept

<<As the students complete the activity in class using the AR/VR headsets, they will fill out an interactive online worksheet that has to be completed along the way. That assignment will count for a grade towards their FSE 100 class.>>

- Our final proposed solution is an interactive AR/VR-based engineering design activity via a worksheet that will allow the FSE 100 students to collaborate with each other, design, and prototype a virtual bookshelf within a simulated environment. The experience integrates the **engineering design process** (identify, research, brainstorm, prototype, test, and communicate) into a fully immersive learning task. Students wear AR/VR headsets to visualize and construct bookshelf models, apply design principles, and iterate on their prototypes while finishing the worksheet without the need for physical materials.
- The solution includes an **interactive online worksheet** (Which will probably be given through canvas) that synchronizes with the VR session. As the students' progress through each design step, prompts and questions appear inside their headset view and on their worksheet screens. They record design decisions, upload screenshots, and document reflections that are automatically saved for grading in their FSE 100 course. The virtual prototype can be modified, rotated, and restructured according to the students' choices. That encourages the principles experimentation, spatial reasoning, and teamwork, all of which are important to FSE 100.
- To effectively engage the students, the activity will feature:
    - A **step-by-step virtual design guide** based on the six-step engineering design process.
    - Real-time **group collaboration tools**, allowing students to co-design the same virtual object from different headset stations.
    - Built-in **rapid prototyping functions**, enabling iteration cycles faster than traditional physical modeling.
    - A **bonus "Design Maze" challenge**, where teams map their decision paths visually within the AR/VR workspace. (May count for extra credit if the professor agrees to give bonus points.)
- This proposed solution connects directly to the FSE 100 course outcomes by blending technical design thinking with emerging technology. The deliverables (worksheet, prototype screenshots, reflection answers) will help students better understand the engineering and design processes.

Artifacts to Convey the Project (Sample Worksheet for Activity below)

FSE 100: Interactive Worksheet - Designing a Virtual Bookshelf

This interactive worksheet guides students through the process of designing and building a virtual bookshelf, applying fundamental engineering principles and exploring the capabilities of AR/VR technologies.

Part 1: The Engineering Design Process

Teams of 3-5 students will engage in the engineering design process to design and virtually build a bookshelf. Students can utilize AR/VR tools or physical blocks for this task.

The 6-step engineering design process is detailed below:

1. **Identify the Problem:** Clearly define the need for the bookshelf and any specific requirements or constraints.
2. **Research and Brainstorm:** Explore existing bookshelf designs, materials, and features. Generate multiple design ideas within your team.
3. **Choose the Best Idea:** Evaluate the brainstormed ideas based on criteria such as functionality, aesthetics, feasibility, and resource availability. Select the most promising design.
4. **Prototype:** Create a virtual model of the bookshelf using AR/VR tools or a physical prototype using blocks.
5. **Test and Evaluate:** Assess the prototype against the initial requirements. Identify any design flaws or areas for improvement.
6. **Communicate Results:** Document the design process, including challenges faced, solutions implemented, and the final design.

Part 2: Rapid Prototyping and VR

Activity A: Limitations of Traditional Rapid Prototyping and VR Solutions

In the table below, list the limitations of traditional rapid prototyping methods and explain how Virtual Reality (VR) addresses these limitations.

Limitations of Traditional Rapid Prototyping	How VR Addresses These Limitations
Limited physical space for prototypes	Unlimited virtual space for diverse prototypes
High material costs for each iteration	Reduced material costs, as iterations are virtual
Time-consuming to build physical prototypes	Faster iteration cycles in a virtual environment
Difficulty in making quick, real-time adjustments	Easy and immediate design modifications

Activity B: Iteration in VR/Block Model

1. Make one significant iteration change to your VR or block model of the bookshelf.
2. Describe a specific design issue identified in the initial prototype.
3. Explain how the iteration improved or resolved this design issue.

Part 3: Communication and Collaboration

This section prompts group discussion on effective communication and collaboration within engineering teams.

- How did your group share and discuss design ideas during the project?
- What challenges did you encounter when trying to explain technical concepts or design decisions to your teammates?
- How can AR/VR tools enhance the way engineers communicate and explain complex designs or technical concepts to others?

Part 4: Bonus Challenge (Optional)

Create a "Design Maze" that visually represents your team's design process for the bookshelf. This can be a hand-drawn diagram or a digital storyboard. The maze should illustrate the different paths, dead ends, and successful routes taken during your design journey.

Part 5: Reflection Questions

Answer the following questions individually:

These pictures are examples of worksheets the students would have to complete after doing one of the VR activities or sitting through one of the lectures.

Patentability

This interactive AR/VR design system can be patentable because it combines education, specifically Engineering, with immersive rapid prototyping. The concept is innovative as it has:

1. **Integration of AR/VR hardware with guided learning modules** for step-based engineering instruction.
2. **Synchronous design documentation system**, where data entered during the virtual activity feeds directly into a live digital worksheet.
3. **Modular adaptability**, allowing the same system to be reprogrammed for other FSE or engineering courses (e.g., bridge design, circuit layout).
    a. Because of its easy-to-use educational interface and real-time feedback loop, this AR/VR worksheet system could be considered for an **educational technology patent** following prototype testing and usability validation.

If successful, our project will enhance the learning experience of engineering concepts to at least 30 students and 2 faculty apart of Arizona State University's FSE 100 class through the usage of virtual reality technology.

3.6 Prototyping

Part 1: Prototyping for Decisions

Research Questions:

- How can a VR game improve the learning process of a student and keep them engaged and active as well as teaching them complex engineering concepts?
    - The answer to this question will help narrow down how we can make the elements provide the most effective learning experience, setting a focus on what the team should target to make an impact.
- How accurate and immersive should this game be to best create a balance between learning and attention to detail?
    - Answering this question will help think of helpful and effective design ideas around graphics and physics. Choosing the right theme will keep the student focused and not distract them too much. It needs to be practical.
- How can the team measure learning outcomes more effectively through VR gameplay. How can this also benefit the instructors?
    - This will help understand what kinds of challenges and virtual lab tasks could best demonstrate educational improvement so we can set a difficulty level that perfectly fits the students' process of learning as well as the instructors' understanding of it. The gameplay should not be too challenging and take a long time to set up.

Part 2: Prototyping Plan

Research Question #1

Research Question:

- How can a VR game improve the learning process of a student and keep them engaged and active as well as teaching them complex engineering concepts?

Varying Variables	VR simulation vs Lecture videos.
Control Variables	Topics covered, Duration of challenge, Number of participants.

Safety Features	Setting the boundary, wireless connectivity, supervision on students.
Constant Variables	Meta Quest VR Headset, lighting, and classroom conditions.
Access to Testing Rig? (Yes/No)	Yes
Creating a Testing Rig? (Yes/No)	No
Obtained Measurement (with units)	Survey (out of 10) about engagement and concept understanding.

Materials:

Material	Quantity
VR headset	20
Compatible VR set up computer.	20
Google forms for surveys	1

Procedure:

1. Develop the unity environment and functionalities for the challenges and learning steps.
2. Create a fully functional syllabus that works together with the practical VR game.
3. The Syllabus must contain brief and clear instructions on what to do for the challenges as well as some notes that work with the videos.
4. VR games must contain helpful brief videos explaining the topics. These videos must be short and entertaining and should keep the students involved.

Results:

Results will be conveyed via a report compiled from various surveys conducted to professors and students, detailing how effective the curriculum and format of the project is. Multiple surveys will be conducted at every stage of the prototyping process, and progress and adjustments will be tracked as well.

Outcomes and Conclusion:

Outcomes will show that VR-based learning environments improve conceptual learning and understanding while being engaging and entertaining compared to traditional methods. This learning method should increase higher retention rates and positive feedback.

Testing Design Requirements:

Test #	Design Requirement	Prototyping Test
1	Students need to be engaged and involved.	Observe user behavior for signs of activity, provide a survey on user experience

2	Students need to have learned or gained further insight of a subject	Test knowledge of users on VR subject with short quiz

Research Question #2

- **Research Question**: **How accurate and immersive should this game be to best create a balance between learning and attention to detail?**

Varying Variables	Low vs Medium vs High Immersion/Detail
Control Variables	Core Learning Objectives and Working Environment
Safety Features	Cleared unobstructed area and short sessions
Constant Variables	Programming, unity structure
Access to Testing Rig? (Yes/No)	No
Creating a Testing Rig? (Yes/No)	Yes
Obtained Measurement (with units)	- Survey 1-10 for student engagement - Short quiz to analyze knowledge retention (%)

Materials:

Material	**Quantity**
VR headset	20
Compatible VR set up computer.	20
Google forms for surveys	1
Google forms for quiz	1
Different Immersion/detailed environments	3

Procedure:

1. Target a class of FSE Students around 20-30.
2. Randomly assign each student to one of the three prototype groups (Low,Medium,High)
3. Briefly introduce the student to the VR controls and safety guidelines.
4. Start the timer and have the student put on the headset and complete the assigned activity.
5. Stop the timer upon activity completion and note the time.
6. Have the student immediately complete the Knowledge Retention Quiz.
7. Have the student complete the User Engagement Survey.
8. Debrief the student and log all results collected.

Results:
Results will be conveyed in comparative data tables and bar graphs. The data from the three groups (Low,Medium,High) will be compiled and compared for each of the four measured metrics (Knowledge %, Engagement Score, Time, Errors). Statistical analysis, like an ANOVA, will be used to determine if observed differences are significant.

Outcomes and Conclusion:
The main takeaway will be a data-driven recommendation for the optimal immersion/detail **level** for the final VR curriculum modules. The takeaway will be a recommendation for a target immersion level that balances learning benefits and practical constraints (performance, cost, sickness). If high detail yields only marginal gains but higher costs, prefer medium.

Testing Design Requirements:

Test #	Design Requirement	Prototyping Test
1	The VR module must effectively teach a core FSE100 concept.	Administer a pre-test and post-test to a group of students using the VR module. A significant improvement in scores validates the requirement.
2	The VR experience must be engaging for students.	Measuring engagement scores (1-10) and observing user behavior for signs of immersion.
3	The software must run stable on the provided VR headsets	Run the VR prototype continuously for a duration exceeding the planned activity length while monitoring for frame rate drops, crashes, or overheating.
4	The user interface must be intuitive and require minimal instruction.	Give students a goal but no instructions on the controls. Observe and record the time to first correct interaction and the number of times they need to ask for help.
5	The activity must be completable within the allocated class time	Ensure the average completion time across all testers is comfortably within the class activity timeframe.

Research Question #3

- **Research Question**: **How can the team measure learning outcomes more effectively through VR gameplay? How can this also benefit the instructors?**

Varying Variables	quiz-only vs gameplay analytics vs mixed-method. automated VR feedback vs instructor-led debrief vs hybrid system. Assessment timing
Control Variables	Same VR activity/module used by all groups. Same learning objectives. class duration.
Safety Features	Clear physical boundaries, guardian mode active. Mandatory breaks.
Constant Variables	Meta Quest VR headset and classroom environment. lighting, room space, and headset configurations.
Access to Testing Rig? (Yes/No)	Yes
Creating a Testing Rig? (Yes/No)	No
Obtained Measurement (with units)	Quiz scores in %. surveys in 1-10

Materials:

Material	**Quantity**
VR headset	20
Compatible VR computers	20
Google Forms surveys	1
VR analytics logging tool	1

Procedure:

1. Develop the analytics system inside Unity to record player actions, errors, hints, and completion time.
2. Introduce each student to the VR module and give a brief explanation of safety rules.
3. Have the student complete the VR activity with analytics enabled.
4. After completion, immediately administer a short quiz
5. Provide instructors with a dashboard output summarizing learning metrics for each student.
6. Compile all results into a comparative dataset and analyze improvement patterns.

Results:

Results will be conveyed via comparative data tables and visualizations

- Gameplay performance vs quiz scores
- Engagement scores vs errors

Outcomes and Conclusion:
The main takeaway will be a validated method for measuring student learning that combines in-game analytics with short quizzes. The instructor will be able to find out which concepts are not getting to students and will allow them to change their teaching patterns accordingly.

Testing Design Requirements:

Test #	Design Requirement	Prototyping Test
1	Learning outcomes must be measurable.	Compare quiz improvement and in-game data to check for patterns in the learning process.
2	VR analytics must accurately track student behavior.	The test logs on multiple students and confirms time, hints, and errors to match real actions.
3	Instructors must be able to interpret the data easily.	Provide instructors with a menu to see all their options and data and get feedback from them.
4	The assessment should correlate with actual concept mastery.	There should be a correlation check between in game performance and quiz

Part 3: Prototyping Updates

Research Question #1

Iteration of Research Question:
How can a VR game improve the learning process of a student and keep them engaged and active as well as teaching them complex engineering concepts?

Proof of Prototyping:

Fractional Capability	Basic VR movement, object interactions (grab, rotate, inspect), and simple physics implemented.

User Interface	brief video popups, and task indicators created.
Systems Capability	Unity with collisions, object interactions, and scripted challenge sequences in a functional unity environment.
Pathfinder	confirmed that the interaction model works smoothly.

Research Question #2

Iteration of Research Question:

How accurate and immersive should this game be to best create a balance between learning and attention to detail?

Proof of Prototyping:

Fractional Capability	Focus on Medium levels of detail in the environment.
User Interface	Different UI styles implemented for each prototype level
Systems Capability	Performance tests and benchmarking
Pathfinder	Students test and give their review

Research Question #3

Iteration of Research Question:

How can the team measure learning outcomes more effectively through VR gameplay? How can this also benefit the instructors?

Proof of Prototyping:

Fractional Capability	Helpful script implemented that tracks completion time, errors, and hints used.
User Interface	Prototype instructor dashboard that displays student performance data.
Systems Capability	VR module stores and prints data to a Google Form.
Pathfinder	Initial test run with 3–5 students to make sure everything works well.

Purchased Materials:

Item Requested	# of Units	Unit Cost	Total Cost	Vendor	Source	Purpose/Justification (please be specific)

Section 4: Detailed Design

4.1 Bill of Materials (B.O.M)

Sub-Assembly	Item	Catalog / Part #	Purchased From	Vendor/ Method	Quantity	Cost Per Unit
/	Meta Quest 3S	/	TBD	TBD	1	$299.99

NOT APPLICABLE - The purpose of this project is to utilize the headsets we were donated, so if something were to break, we will make do with the remaining headsets instead of buying more.

4.2 Prints/Schematics/Code

Semester	Part	Description of Part / Links
Fall 2025	Environment	The environment is programmed and designed in Unity where the student will start in the VR headset. The student can maneuver and manipulate the environment around them
Fall 2025	Interactables	The interactables are what allow the objects and components in the program to interact with each other
Fall 2025	Assets	The assets are the different components of the space such as the tables, materials, and objects that are included in the environment

4.3 Manufacturing and Assembly Processes

Part 1: Creation of Virtual Environment

Materials:

Unity.

Procedures:

1. Design a simple 3D environment within unity for the experience to occur within.
2. Create components that will be used to create the prototypes within the experience.
3. Design tools that the students will use within the environment

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Part 2: Implementation of Physics and Mechanics

Materials:

Unity, Visual Studio Code

Procedures:

1. Implement physics simulations within the experience.
2. Design iterative design specification mechanics, and verification system.
3. Design functionality to save and load designs.

Part 3: Creation of Curriculum and Teacher Resources

Materials:

Video recording and editing software, text editing software.

Procedures:

1. Record video guides for instructors to know how to handle and operate the headsets.
2. Document common issues and troubleshooting solutions in a user's manual.
3. Document the process of the experience for instructors.

4.4 Risk Analysis

Development Process Risks

- **Incomplete or buggy software due to limited Unity/VR experience**
  Impact: Students may experience glitches, motion issues, or confusion, lowering learning quality
  Mitigation: Iterative testing with small user groups, simpler mechanics, frequent faculty feedback
- **Unclear Requirements**
  Impact: Features may be incomplete or inconsistent by delivery, reducing instructional value
  Mitigation: Set clear MVP, lock requirements early, review frequently with Professor Osburn

Design Risks

- **Overly complex VR interactions or unclear UI**
  Impact: Students may become frustrated or unable to complete the activity, reducing engagement
  Mitigation: Use simple controls, minimal menus, and conduct usability tests with non-VR users
- **Graphics or motion not optimized**
  Impact: Can cause nausea, VR sickness, or inability to finish the activity
  Mitigation: Maintain stable frame rate, limit head-movement intensity, add comfort settings

Prototype / Hardware Risks

- **Headsets may not be fully charged, updated, or functional during class**
  Impact: Students may not get time with the headset, activities fall behind schedule.
  Mitigation: Pre-class maintenance checklist, TA responsibility assignments, backup devices
- **Physical safety issues**
  Impact: Risk of minor injury or equipment damage
  Mitigation: Define VR-safe zone in classroom, require seated/standing-still interactions, TA supervision

4.5 Verification

Spec #	Design Requirement	Verification
1.1	*The VR program must act in the manner of a video game that relates to the curriculum so that the professors can relay instructions and goals of the program. The program will have 3 – 4 levels and colorful assets to be more like a video game. The professor's instructions will include how to use the VR headsets to pick*	Conduct a sample test with at least 5 students and one instructor to ensure that the interface functions like a video game (including levels, goals, and clear user interactions). Confirm with instructor feedback that the game structure aligns with the curriculum goals. We want it as a video game because we want to test for

	up items, build items, and design prototypes the students come up with.	engagement. We will survey the professors to learn the engagement differences between a normal lecture and this AR/VR activity.
1.2	*VR programs must utilize user input (such as picking something up or moving object around while wearing the headsets) where the user can maneuver and control the environment around him/her, as well as put the user in missions with goals needed to be accomplished. In addition, there should be graphics that are interesting and aesthetically relevant and pleasing.*	Perform in-class testing to confirm that motion tracking, object manipulation, and environmental controls function correctly. Collect user feedback on motion accuracy, ease of interaction, and visual satisfaction through a short post-test survey.
1.3	*In addition, there should be graphics that are interesting and aesthetically relevant and pleasing. In the post event surveys, the average response for the question "This program had good graphics and scenes" should be 6-10.*	Distribute a post-session survey using a 1–10 Likert scale to at least 20 participants. Calculate the average score to verify that the overall aesthetic rating is 6 or above.
2.1	The curriculum should be able to accommodate every student being able to use the headsets at least one. If there are enough headsets, one can be given to each student, otherwise the course should be adaptable to allow students to take turns or share within a group.	Plan: Run a practice simulation of the VR activity with different amounts of headsets and check if the activity is still able to be done by the students. If a certain number of headsets is too little, make sure we always have more headsets than that amount. Conduct a trial class session with varying headset numbers (e.g., 4, 6, 8) to determine minimum requirements for full participation. Verify that all students are able to engage in the activity within a single class session.
3.1	*This VR project needs to be well made and informative at a moderate level so that the FSE and education faculty can see it as a helpful source of teaching students*	Have one or more faculty members and at least three students review the module and provide qualitative feedback on clarity, pacing, and

	through newer methods. The lessons in the curriculum should be beginner friendly and accommodate extra time in case of technical difficulties.	difficulty level. Confirm that the module can be completed within one class period even if 10–15 minutes are lost to setup or troubleshooting.
3.2	The slides should include why VR prototyping is useful for students. At least 75% of the students should be able to write about how they can use VR prototyping in their designing processes.	Try to get a short reflection at the end of the session. Verify that at least 75% of the responses correctly describe one or more applications of VR prototyping in engineering design.
3.3	*The lessons in the curriculum should be beginner friendly and accommodate for extra time in case of technical difficulties. The class should be able to go through one whole activity within the time limit for one class.*	Time a full FSE 100 class session from setup to completion to ensure that students can complete the task within 50–75 minutes. Record whether additional time is needed due to any potential technical delays.
4.1	At least 75% of the students should leave the course understanding what AR/VR is and how to use the headsets at least at a basic level.	Plan: Create a quiz that covers all materials in the slides. Make sure using the slides students are able to answer all the questions Created post-lesson quizzes covering VR fundamentals and headset operation. Make sure that at least 75 – 80% of students achieve a score of 70% or higher.
5.1	All VR headsets and controllers must follow campus health & safety guidelines. This means proper sanitation between uses and safe fit to avoid strain. Equipment should be inspected weekly for damages or malfunctions, and a	Maintain a written or digital inspection checklist and weekly log. Verify that all equipment passes safety inspection and sanitation check before each class session.

	maintenance log must be kept to ensure devices remain in working condition.	

4.6 Validation

Need #	User Need	Validation
1.1	*The VR program must act in the manner of a video game that relates to the curriculum so that the professors can relay instructions and goals of the program.* *The program will have 3 – 4 levels and colorful assets to be more like a video game. The professor's instructions will include how to use the VR headsets to pick up items, build items, and design prototypes the students come up with.*	Run a sample test run from students and record whether they can successfully navigate levels, follow instructions, and complete required interactions without instructor or TA assistance. Collect feedback to confirm the experience feels game-like, engaging, and aligned with the curriculum goals.
1.2	*VR programs must utilize user input (such as picking something up or moving object around while wearing the headsets) where the user can maneuver and control the environment around him/her, as well as put the user in missions with goals needed to be accomplished. In addition, there should be graphics that are interesting and aesthetically relevant and pleasing.*	Observe all actions during a live demo session and confirm that all required user interactions (pick up, move, manipulate objects) function correctly. Also collect a short survey for the ease of interaction and visual quality to ensure user input is intuitive, easy to navigate, and that the graphics meet expectations.
1.3	*In addition, there should be graphics that are interesting and aesthetically relevant and pleasing. In the post event surveys, the average response for*	Create and give a survey after the VR experience to all participating students

	the question "This program had good graphics and scenes" should be 6-10.	and calculate the average rating they gave for graphics. Acceptance criteria are met if the mean score is ≥ 6.
2.1	The curriculum should be able to accommodate for every student being able to use the headsets at least one. If there are enough headsets, one can be given to each student, otherwise the course should be adaptable to allow students to take turns or share within a group.	Plan: observe a class doing the VR activity and see if every student is learning and getting to use the VR headsets despite varying amounts of headsets. Monitor the activity throughout class time and potentially have an attendance/participation log or checklist to make sure that all students were successfully able to use the headset regardless of the total headset quantity.
3.1	*This VR project needs to be well made and informative at a moderate level so that the FSE and education faculty can see it as a helpful source of teaching students through newer methods. The lessons in the curriculum should be beginner friendly and accommodate for extra time in case of technical difficulties.*	Conduct faculty review from all FSE 100 professors (if possible) and a preview class test. Faculty should confirm instruction clarity, and students should be able to complete the module within the allotted class time even if there is a potential 10–15-minute technical delay.
3.2	The slides should include why VR prototyping is useful for students. At least 75% of the students should be able to write about how they can use VR prototyping in their designing processes.	Distribute a brief reflection prompt and survey after the lesson and total up the responses. Requirement will be met if ≥75% provide correct explanations demonstrating

		understanding of VR prototyping benefits.
3.3	*The lessons in the curriculum should be beginner friendly and accommodate for extra time in case of technical difficulties. The class should be able to go through one whole activity within the time limit for one class.*	Run a timed classroom test and document whether students complete the full activity within one class period (50–75 minutes). Success is confirmed if the full workflow is completed despite delays.
4.1	At least 75% of the students should leave the course understanding what AR/VR is and how to use the headsets at least at a basic level.	Plan: Make sure that without the lecture slides, at least 75% of the students can pass a post-lecture quiz.
5.1	All VR headsets and controllers must follow campus health & safety guidelines. This means proper sanitation between uses and safe fit to avoid strain. Equipment should be inspected weekly for damages or malfunctions, and a maintenance log must be kept to ensure devices remain in working condition.	Plan: contact ASU health and safety services or dreamscape and interview them on the necessary safety precautions we need to take when using VR headsets. Practice these practices then have one of them observe us and rate us from 1-10 on our safety quality.

Section 5: Project Delivery

5.1 User/Service Manual

As of now, there is no complete user manual. We are still working on making our virtual environment design functional. Furthermore, we need to finalize our draft for the curriculum to make an activity for the students to follow. Once we have the curriculum and the virtual environment designed, we will make an instruction manual for the students to follow and complete their activity.

I am in the process of creating the curriculum with lectures, integrating with Unity AR/VR, and creating quizzes that the students can take. Once the curriculum is finished and developed, we will create user manual for students and teachers. The following is the link to the progress my team and I have made for the curriculum section. curriculum folder

5.2 Delivery Checklist

Product, Process, or System Detailed Schematic, or Workflow (For Coding)

Once we have a finalized design to build, we will provide the process to which we are building our functional working environment.

Bill of Materials / Software or Frameworks (For Coding)

We are still in the planning stages of our project and have just started to learn the Unity software platform. Once we start building the final design for our virtual environment, we will provide the bill of materials/software or frameworks.

Functional and Usability Test Results

We haven't been able to run any tests yet, so we do not have any available functional and usability test results.

Safety and Reliability Test Results

We haven't been able to run any tests yet, so we do not have any available safety and reliability test results.

System and Field Test Results

We haven't been able to run any tests yet, so we do not have any available field test results.

User and Training Manuals

We are still in the planning stages of our project, so we will provide the user and training manuals once we have a functional virtual environment design.

Community Partner Evaluation of Final Design Results

Once we finalize our functional virtual environment design, we will provide the result document link.

5.3 Customer Satisfaction Questionnaire